

MIAE 221 X Materials Science Department of Mechanical, Industrial and Aerospace Engineering Fall 2026				
Course Instructor: Dr. Mamoun Medraj Email: mamoun.medraj@concordia.ca				
Office Hours:	Wednesdays	03:00 pm – 05:00 pm	EV 12.185	
Lectures:	Wednesdays & Fridays	08:45 am – 10:00 am	H 937	
Tutorials: Please see your class schedule for details				
Section	Day	Time	Classroom	TA
XA	Friday	2:45pm–3:35pm	H 564	Ashenafi Bicks
XB	Friday	2:45pm–3:35pm	H 562	Lama Mahmoud
XC	Monday	5:45pm–6:35pm	CL 235	Shabnam Moradi
XD	Friday	4:15pm–5:05pm	CL 235	Lama Mahmoud
XF	Wednesday	2:45pm–3:35pm	H 561	Michael Pudlo
Course Calendar Description: <i>Relationships between properties and internal structure, atomic bonding; molecular, crystalline and amorphous structures, crystalline imperfections and mechanisms of structural change. Microstructures and their development from phase diagrams. Structures and mechanical properties of polymers and ceramics. Thermal, electrical and optical properties of materials. Lectures: three hours per week. Tutorial: one hour per week.</i>				
Prerequisites: CHEM 205 (CEGEP Chemistry 101) Co-requisites: N/A This course is a prerequisite to: MECH 321 & AERO 481				
Specific Knowledge and Skills Needed for this Course: Students taking this course are expected to have sufficient knowledge of the following topics. Should you have difficulties in any of these topics, you are strongly encouraged to review them before the DNE deadline. <i>Atomic structure, electricity, light, chemical reactions.</i>				

Course materials

Required Textbook: “*Materials Science and Engineering: An Introduction*” 10th Edition, Callister & Rethwisch, Wiley.

The current edition is also available as an ebook through the Bookstore or Wiley (www.wileyplus.com). You can purchase the ebook or subscribe to the WileyPlus experience which has additional resources in addition to the text.

A hardcopy of the 5th - 9th Edition will also work.

Related Material: There are other texts and reference material in the Library on all aspects of Engineering Materials if you require further information or a different explanation. Concordia University Library: TP1120 - Plastics, TP800 – Ceramics, TA400 – Materials, TA418.9 – Composites.

Grading Scheme

<i>In-Tutorial Assignment Problems</i>	20 %
<i>Midterm</i>	25 %
<i>Term Assignment</i>	10 %
<i>Final Exam (3 hours closed book)</i>	45 %
<i>Total</i>	100%

Passing Criteria:

In order to pass the class, both your cumulative score and the final examination score must be above 50%.

Tentative Course Schedule

Topic	Week	Chapters
<i>Introduction to Materials Science</i>	<i>Week 1</i>	<i>Ch 1</i>
<i>Atomic Structure & Interatomic Bonding</i>	<i>Week 1 & 2</i>	<i>Ch 2</i>
<i>The Structure of Crystalline Solids</i>	<i>Week 2 & 3</i>	<i>Ch 3</i>
<i>Imperfections in Crystals</i>	<i>Week 4</i>	<i>Ch 4</i>
<i>Diffusion</i>	<i>Week 5</i>	<i>Ch 5</i>
<i>Mechanical Properties of Metals</i>	<i>Week 6</i>	<i>Ch 6</i>
<i>Dislocations & Strengthening Mechanisms</i>	<i>Week 7</i>	<i>Ch 7.1 – 7.4</i>
<i>MIDTERM EXAM</i>		
<i>MIDTERM Break</i>		
<i>Phase Diagrams</i>	<i>Week 8</i>	<i>Ch 9</i>
<i>Structures, Properties of Ceramics</i>	<i>Week 9</i>	<i>Ch 12.3 - 12.11 & 13.11</i>
<i>Polymer Structures and properties</i>	<i>Week 10</i>	<i>Ch 14 & 15.1 - 15.14</i>
<i>Electrical, Thermal, and Optical Properties</i>	<i>Week 11 & 12</i>	<i>Ch 18 & Ch 19 & Ch 21</i>
<i>Review</i>	<i>Week 12</i>	

Details on assessment tools:

1. **Tutorial Problems:** Sample questions will be posted on the web page. Solutions will be covered in tutorials. They will NOT be marked but these questions will be used as the basis for the In-Tutorial Exercises which will take place every second tutorial and will be marked.

2. **Midterm Examination:** The date for the midterm has been tentatively scheduled for **FRIDAY, October 30th** during lecture time.

3. **Term Assignment:** Individually, or as a team of two, research a certain assigned topic pertinent to the material covered in this course. The report, in your own words, should be between 1500-2000 words and should include references and diagrams where appropriate. The topic will be posted in a separate document. The due date for this Term Assignment is **November 27th** but it can be submitted earlier.

Other information

REASONING

The reason why knowledge of materials is useful to you is that all engineers make use of materials in one form or other whether they are designing gas turbine engines, cell phones, wind turbines or an artificial hip joint and the ability to make the correct selection of material for a particular application or to understand the response of a material to an applied condition, is of paramount importance. Choosing a suitable material for an application is a demanding task as mistakes can be costly in many ways, and an understanding of the structure and behaviour of materials are essential to go beyond the "well, this is what they used last time" approach.

LECTURES

You are strongly encouraged to attend all lectures as this will allow you to ask questions and gauge the relevant importance of topics. Lecture slides are available in .pdf format on the Moodle site.

OFFICE HOURS

I will be available during Office hours to answer specific problems on the course material. You should contact your Tutor for questions related to tutorial assessment grading.

Graduate Attributes:

The following is the list of graduate attributes (skills) that students use, learn and/or apply throughout the term.

A knowledge base for engineering.

Knowledge-base of natural science

Life-long learning

Identifying missing knowledge and self-learning.

Course Learning Outcomes (CLOs):

By the end of this semester, students are expected to master the following concepts.

Course Learning Outcomes

Related Graduate Attributes

Describe the importance of atomic bonding on the structure and behaviour of engineering materials.	A knowledge base for engineering/ Knowledge-base of natural science
Demonstrate a knowledge and understanding of the different material systems, their structures and physical properties. Explain diffusion behaviour in solids through a knowledge of atomic structure and laws of diffusion.	
Demonstrate an understanding of the basic mechanical behaviour of materials and explain the difference between strength and stiffness.	
Use phase diagrams to describe alloy microstructures and calculate expected phase compositions and proportions, in particular in the Fe-Fe ₃ C system.	
Explain how the bonding and atomic structure of the different materials affects their thermal, electrical and optical behaviour.	
Find out relevant material from various sources to prepare a summary report on a topic related to but not directly covered in class.	Life-long learning
Health and Safety Guidelines General health and safety instructions and available health and safety trainings can be found at: Safety Programs - Concordia University (https://www.concordia.ca/campus-life/safety/general-safety.html)	
On Campus Resources Please visit Student services at Concordia University for the services available Gina Cody School students.	
Disclaimer In the event of extraordinary circumstances beyond the University's control, the content and/or evaluation scheme in this course is subject to change.	
ACADEMIC CONDUCT ISSUES THAT APPLY IN GENERAL The basic rules that make you a good engineer The B. Eng. program is set to satisfy most of the requirements for your education and prepares you for a professional engineering career that requires dedication and knowledge. What you learn, and how you learn, will be used extensively in your engineering profession for the next 30 to 40 years. Therefore, the four years spent in the engineering program are crucial towards your professional formation. The first step is for you to learn to “think like an engineer” which means: • accept responsibility for your own learning • follow up on lecture material and homework	

- learn *problem-solving skills*, not just how to solve each specific homework problem
- build a body of knowledge integrated throughout your program
- behave responsibly, ethically and professionally

One of the mainstays of being a professional engineer is a professional code of conduct and as an engineering student this starts with the Academic Code of Conduct (Article 16.3.14 of the undergraduate calendar). However, you may encounter situations that fall outside the norm and in such cases, you use your common sense.

Further, the following issues should be given serious consideration:

- 1) Attendance at lectures and tutorials are major learning opportunities and should not be missed. Class and tutorial attendance is important for you to comprehend the discipline and make the connections between engineering skills. You are strongly encouraged to participate in the class, ask questions and answer the instructor's questions. Tutorials are just extensions of the classes in which application of the concepts presented during the lectures are presented and problems are practically solved.
- 2) Homework is usually mandatory and it has some weight in the final grade (such information is given in the course outline). Homework may also be conceived as training material for the class tests. Under all circumstances, it is highly recommended to carry out the home work on time and submit it on the prescribed date. Late submissions are not granted to individual cases regardless of the reason. This is part of the training for being in the workforce where deadlines have to be met. Please, plan your work such that you submit all the assignments on time and in the correct place (not in the corridor or on the street!).
- 3) You will be informed of office hours with tutors or class instructors. Please respect these office hours and in case you have a serious conflict, contact the instructor asking for a special time arrangement.
- 4) Class tests (midterms, quizzes) are returned to the student. The final exams are not. If you wish to see your exam paper, be aware that most instructors allow only a narrow window of time for that purpose. For the fall term, exams may usually be reviewed in January and May for the spring term.
- 5) When you see your marked work (assignments, midterms, final exam etc), be aware that you are supposed to review your material and see the type of errors you made and if marks have been added incorrectly. This is not an opportunity to try and "negotiate" a higher grade with the instructor. If you believe that your grade is not right, you may apply for a formal Course Reevaluation through the Birks Student Centre.
- 6) Writing tests and exams represents a major component of your course work. These tests and exams have rigorous requirements such as:
 - Only **specified faculty calculators** are allowed during tests and exams unless otherwise indicated by the instructor. Usually, **no materials** are allowed in the exam unless otherwise announced.
 - **Get used to signing in and out of MOODLE and try the practice exams.**
- 7) Respect your colleagues and those that you meet during the class: tutors, instructors, teaching assistants, etc. Use appropriate communication means and language. Be considerate for all human beings. This includes small things such as turning off cell-phones before a class begins. Concordia University is a very diverse group of people and a very large multicultural community.
- 8) Communication is part of your future profession. Learn how to communicate effectively and efficiently in the shortest time possible. Write short but meaningful e-mails, make effective phone

calls, etc. If your instructor accepts emails make sure that your request is clear with the course number and your name in the *Subject* line. Do not ask for special treatment as instructors have to treat all students equitably.

Respect all the above and you will get closer to your future profession.